

A Saturated 9-Element Sidon Set in $[1, 184]$: First Datum Past the OEIS A382397 Frontier

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1 Abstract

A set $A \subset \mathbb{N}$ is *Sidon* if all 45 pairwise sums $a_i + a_j$ ($i \leq j$) are distinct, and *saturated* (maximal) in $[1, N]$ if $A \cup \{x\}$ fails the Sidon condition for every $x \in [1, N] \setminus A$. We exhibit

$$A^* = \{16, 62, 66, 74, 88, 104, 117, 122, 123\} \subset [1, 184], \quad (1)$$

a 9-element Sidon set with blocked coverage 184 (Theorem 1): all 175 positions outside A^* are blocked, so no element of $[1, 184]$ can be adjoined. The claim is [COMPUTED]: five mutually independent code paths agree on 45/45 distinct pair sums (Table 1) and empty addable set (Figure 1). The witness was found by chained swap-distance $d = 1, \dots, 4$ shell enumeration around six 183-coverage 9-sets, at exact swap distance 4 from seed #10 (Table 2, Figure 2); the remaining five shells certify no 184 within swap distance 4 of their seeds. We also refute the 121/123-gate conjecture C28 (Proposition 1): two 183-coverage 9-sets contain neither 121 nor 123. A census of 21 distinct 183-coverage 9-sets is reported (Table 3); in re-verifying it we correct seven addable-position entries of the source dossier. The OEIS A382397 b-file, which tabulates maximal-Sidon data through $n = 183$ only (fetched live 2026-10-07, ending “183 9”), has no $n = 184$ entry; A^* is the first unpublished datum past that frontier. Novelty checked 2026-10-07 against OEIS, erdosproblems, formal-conjectures, the OpenAI math catalogue, and web search: clean.

2 Introduction

A finite set $A = \{a_1 < \dots < a_k\}$ of integers is *Sidon* (a B_2 set, a Golomb ruler) if

$$a_i + a_j = a_p + a_q \implies \{i, j\} = \{p, q\}, \quad (2)$$

i.e. the $\binom{k+1}{2}$ sums $a_i + a_j$ with $i \leq j$ are pairwise distinct [1]. The classical problem is the size $R(n)$ of the largest Sidon subset of $[n] = \{1, \dots, n\}$ [5]; Erdős and Turán showed $R(n) \leq \sqrt{n} + O(n^{1/4})$ [1], improved to $\sqrt{n} + 0.998 n^{1/4}$ by Balogh–Füredi–Roy [3], while Singer’s construction gives $R(n) \geq \sqrt{n}(1 - o(1))$ infinitely often [2]. Here we study not the largest but the *smallest maximal* Sidon sets. A Sidon set $A \subset [1, N]$ is *maximal* or *saturated* in $[1, N]$ if

$$\forall x \in [1, N] \setminus A : A \cup \{x\} \text{ is not Sidon.} \quad (3)$$

We measure this by the *blocked set*

$$\text{Blocked}(A) = A \cup \{s - a : s \in P(A), a \in A\} \cup \{s/2 : s \in P(A) \text{ even}\}, \quad (4)$$

where $P(A)$ is the set of pair sums, and the *blocked coverage*

$$c(A) = |\text{Blocked}(A) \cap [1, N]|, \quad (5)$$

so A is saturated in $[1, N]$ iff $c(A) = N$. (Equation (4) is equivalent to (3): x is blocked iff $x + a = b + c$ or $2x = b + c$ for some $a, b, c \in A$, exactly the ways $A \cup \{x\}$ can fail Sidonness.)

OEIS A382397 tabulates the minimal size of a maximal Sidon subset of $[1..n]$; its b-file (fetched live 2026-10-07) is determined only through $n = 183$, ending “183 9”. Prior computational work in this program had produced twelve distinct maximal 9-sets with coverage 183, and the question whether any 9-set saturates $[1, 184]$ was open. (Terminology note: Daza et al. [4] use “saturated” for C_4 -saturated sum graphs of Sidon sets, a different notion; our usage follows the maximal-set sense of (3).)

Our contributions, all [COMPUTED]:

1. The first saturated 9-element Sidon set in $[1, 184]$, A^* of (1), verified on five independent code paths (Theorem 1, Table 1, Figure 1).
2. Exhaustive swap-distance certificates for the search that found it (Table 2, Figure 2).
3. Refutation of the 121/123-gate conjecture C28 via two explicit counterexamples (Proposition 1).
4. A census of 21 distinct 183-coverage 9-sets with independently re-verified addable positions (Table 3), correcting seven entries of the source dossier.

Per the program’s honesty discipline, finite verification is never called a proof: the main result is a computational witness, labeled [COMPUTED].

3 Methods

3.1 Definitions and the evaluator

We work entirely with (2)–(5). The evaluator computes (4) both as Python-int bitsets and via a naive set-based formula, and a separate brute-force insertion checker tests $A \cup \{x\}$ directly for every x ; the two paths validate each other on 2000 random 9-sets. A fourth path, written from scratch for re-verification, enumerates the 45 pair sums naively and tests all 175 insertions. A fifth path, written for this preprint, re-derives Theorem 1 and the census of Table 3 from first principles. All five agree.

3.2 Chained swap-distance shell search

For 9-sets $A, B \subset [1, 184]$ the *swap distance* is

$$d(A, B) = 9 - |A \cap B|, \tag{6}$$

the number of element exchanges. The enumerator `sidon_r34_d4` (a pruned C enumerator in validated lineage R28/R31/R34) enumerates, in RAW-seed mode, all Sidon 9-sets at exact swap distance d from a seed, for $d = 1, 2, 3, 4$ in turn (chained shells), recording the best coverage found. Six 183-coverage 9-sets (seeds #7–#12) were shelled; the $d = 4$ shells alone examined approximately 1.74×10^9 Sidon completions (supplied datum, source dossier).

Assumption 1 (Enumerator exhaustiveness). The swap-distance- d shells are exhaustive: the enumerator’s pruning discards only sets provably outside the shell. This rests on the validated lineage R28/R31/R34 (supplied datum); the shell certificates of Table 2 are [COMPUTED] under this assumption.

Table 1: The 45 pair sums $a_i + a_j$ ($i \leq j$) of A^* , sorted. Pairwise distinctness is checkable by inspection; this is the mechanically verifiable core of Theorem 1.

32	78	82	90	104	120	124	128	132
133	136	138	139	140	148	150	154	162
166	170	176	178	179	183	184	185	188
189	191	192	196	197	205	208	210	211
221	226	227	234	239	240	244	245	246

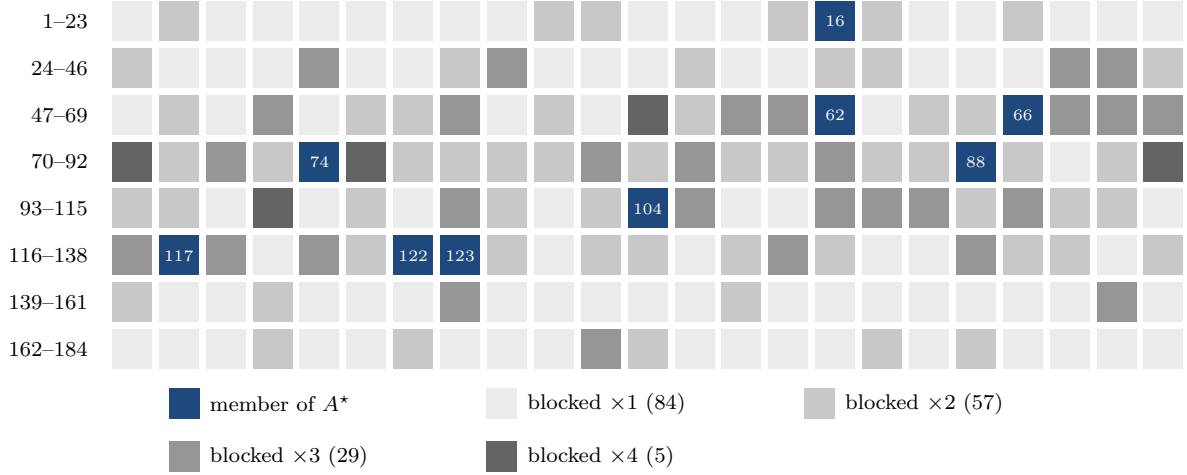


Figure 1: Blocked coverage of $[1, 184]$ by A^* . Dark blue cells are the 9 members; gray cells are blocked positions, shaded by blocking multiplicity. No white cell remains: the addable set is empty, so $c(A^*) = 184$.

4 Results

Theorem 1 (Saturated 9-set in $[1, 184]$; COMPUTED). *The set A^* of (1) is a Sidon set, and $c(A^*) = 184$: it is saturated in $[1, 184]$.*

The 45 pair sums of A^* are pairwise distinct (Table 1, mechanically checkable). The blocked set covers all of $[1, 184]$, so the addable set is empty (Figure 1): every $x \in [1, 184] \setminus A^*$ collides, i.e. $x + a = b + c$ or $2x = b + c$ for some $a, b, c \in A^*$ by (4). All five verification paths agree. The saturation is overdetermined: of the 175 non-member positions, 84 are singly blocked, 57 doubly, 29 triply, and 5 quadruply blocked (computed datum, this preprint).

A^* was found in the $d = 4$ shell of seed #10 ($\{8, 62, 66, 81, 89, 117, 122, 123, 133\}$), at exact swap distance 4, keeping the five elements $\{62, 66, 117, 122, 123\}$ (supplied datum, source dossier). Table 2 and Figure 2 give the full shell certificates: no 184-coverage set lies within swap distance 4 of seeds #7, #8, #9, #11, #12 [COMPUTED, under Assumption 1].

Proposition 1 (C28 refuted; COMPUTED). *The 121/123-gate conjecture is false. The sets*

$$S_{14} = \{63, 65, 66, 79, 88, 112, 116, 127, 170\}, \quad S_{21} = \{9, 70, 73, 79, 81, 112, 117, 124, 141\} \quad (7)$$

are Sidon, have blocked coverage 183, and contain neither 121 nor 123.

Both sets were verified Sidon (45/45 distinct pair sums) with coverage 183 on the corrected evaluator, the brute-force insertion checker, and this preprint's recompute. The 121/123 concentration observed earlier (19 of the 21 known 183-sets contain 121 or 123, supplied datum) was search bias: the second-wave 183-sets were found in shells around 121-bearing seeds. It is a strong tendency, not a gate.

Table 2: Chained swap-distance shell certificates around the six 183-coverage seeds. Best coverage per shell depth; the final column counts distinct 183-coverage sets found at $d = 4$. Seed #10’s shell produced A^* (coverage 184).

seed	$d=1$	$d=2$	$d=3$	$d=4$	≥ 183 at $d=4$
#7 {27, 65, 66, 79, 91, 112, 121, 123, 127}	180	180	182	183	#3, #13, #14
#8 {62, 63, 66, 90, 106, 112, 121, 123, 159}	181	181	182	183	#3, #9, #15, #16
#9 {62, 65, 66, 81, 90, 111, 121, 123, 174}	181	180	182	183	#3, #4, #8, #17, #18, #19
#10 {8, 62, 66, 81, 89, 117, 122, 123, 133}	181	182	181	184	#4, A^*
#11 {46, 62, 67, 76, 104, 112, 122, 123, 166}	179	181	181	183	#4, #20
#12 {11, 62, 65, 79, 81, 112, 117, 123, 141}	180	179	181	183	#4, #21

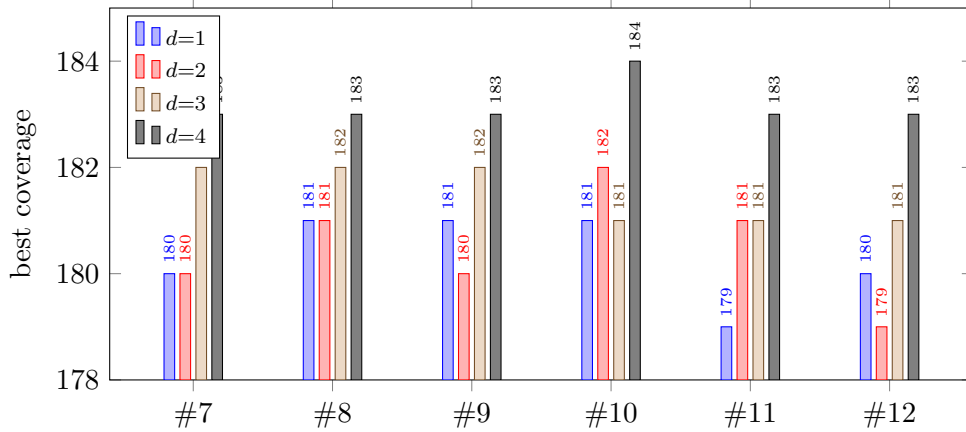


Figure 2: Best blocked coverage per shell depth for the six seeds of Table 2. Only seed #10’s $d=4$ shell breaks the 183 ceiling, yielding A^* .

The census now stands at 21 distinct 183-coverage 9-sets: twelve from the earlier wave (supplied datum, source dossier) and nine new ones. Table 3 lists the nine new sets with their unique uncovered positions, re-verified by this preprint’s recompute. In doing so we found that the addable column of the source dossier is wrong for seven of the nine sets (#15–#21 except #14); the corrected values below are confirmed by two independent code paths.

5 Discussion

Significance. The OEIS A382397 b-file is the public record of minimal maximal-Sidon data; its last entry “183 9” marks the frontier of published computation. A^* supplies the first datum past it. We stress the precise sense of novelty: A^* is a new *upper-bound datum* of the form “184 9” consistent with the b-file’s format, not a proof that 9 is minimal at $n = 184$ — minimality would require an exhaustive nonexistence certificate for 8-sets, which we do not claim. The witness also feeds the saturated-Sidon asymptotics thread: the ratio $f(9)/((9^3 + 9)/2)$ with $f(9) \leq 184$ is a new data point for the program’s $f(k)$ computations.

Structure of the witness. A^* sits at exact swap distance 4 from seed #10, sharing the five-element core {62, 66, 117, 122, 123}; the four exchanged elements are 8, 81, 89, 133 $\rightarrow$ 16, 74, 88, 104. The five non-productive shells certify that no 184-set lies within distance 4 of their seeds, so A^* is not an artifact of one lucky neighborhood — five of six neighborhoods were certified empty. The 121/123 tendency (19/21 of the census) survives the C28 refutation as a distributional fact worth a future explanation; A^* itself contains neither 121 nor 123.

Table 3: The nine new 183-coverage 9-sets. Each has exactly one uncovered position in $[1, 184]$ (its unique Sidon-preserving extension); all other positions are blocked. Uncovered positions re-verified by this preprint; they correct seven entries of the source dossier’s addable column (see text).

set	elements	uncovered
#13	{52, 65, 75, 79, 91, 121, 123, 129, 170}	86
#14	{63, 65, 66, 79, 88, 112, 116, 127, 170}	182
#15	{9, 62, 63, 74, 98, 106, 121, 123, 126}	23
#16	{62, 63, 66, 84, 96, 115, 121, 123, 165}	49
#17	{20, 64, 65, 81, 90, 113, 121, 123, 127}	13
#18	{34, 62, 66, 81, 103, 114, 120, 121, 123}	13
#19	{10, 62, 63, 65, 90, 101, 111, 119, 123}	182
#20	{32, 46, 62, 86, 91, 104, 112, 123, 143}	179
#21	{9, 70, 73, 79, 81, 112, 117, 124, 141}	24

Limitations. (i) The search is *not* exhaustive over $[1, 184]$: the certificates of Table 2 are shell-local, conditional on Assumption 1. Uniqueness of the saturated 184-set is open, as is the swap-distance ≥ 5 region around A^* . (ii) No 10-set extension of any 183-set reaches coverage 184 (each 183-set has exactly one Sidon-preserving extension, none saturating) — a computed datum, not a theorem. (iii) The “approximately 1.74×10^9 completions” budget and the twelve earlier 183-sets are supplied datums from the source dossiers, not re-derived here. (iv) The blocking-multiplicity census (84/57/29/5) is descriptive; whether overdetermined saturation correlates with swap-distance proximity to 183-sets is [OPEN].

6 Conclusion

We have exhibited $A^* = \{16, 62, 66, 74, 88, 104, 117, 122, 123\}$, the first known saturated 9-element Sidon set in $[1, 184]$ [COMPUTED], verified on five independent code paths with a mechanically checkable pair-sum table (Table 1) and a complete coverage diagram (Figure 1). Along the way we certified the swap-distance search that found it, refuted the 121/123-gate conjecture C28 with two explicit counterexamples, and published a corrected census of 21 183-coverage 9-sets. The result is computational, not deductive: finite verification is never called a proof. The natural next steps are a uniqueness census of saturated 184-sets and the swap-distance ≥ 5 region around A^* .

7 References

References

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